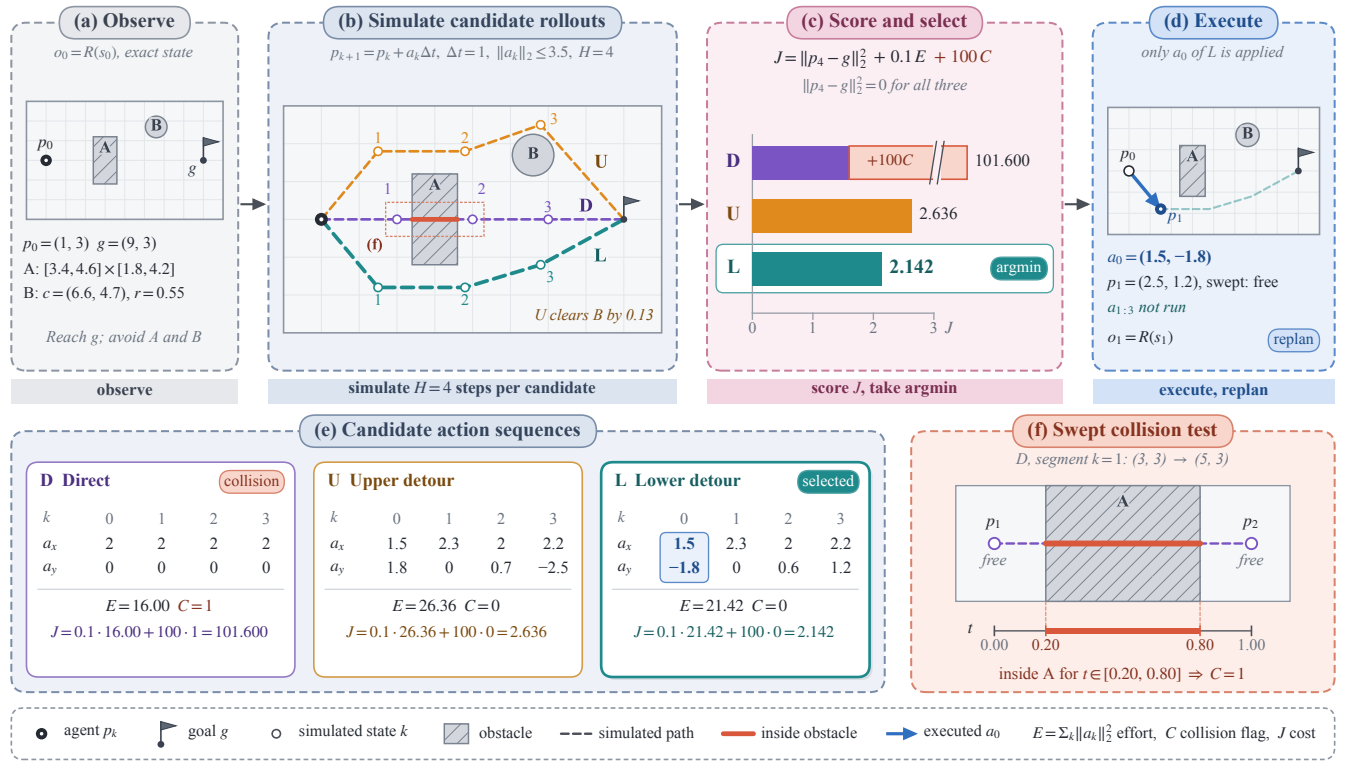

(a) Observe
$o_0 = R(s_0)$, exact state
p_0
A
B
g
$p_0 = (1, 3)$ $g = (9, 3)$
A: $[3.4, 4.6] \times [1.8, 4.2]$
B: $c = (6.6, 4.7)$, $r = 0.55$
Reach g; avoid A and B
observe
(b) Simulate candidate rollouts
$p_{k+1} = p_k + a_k \Delta t$, $\Delta t = 1$, $\|a_k\|_2 \le 3.5$, $H = 4$
U
D
L
(f)
U clears B by 0.13
simulate $H = 4$ steps per candidate
(c) Score and select
$J = \|p_4 - g\|_2^2 + 0.1\,E + 100\,C$
$\|p_4 - g\|_2^2 = 0$ for all three
D
+100C
101.600
U
2.636
L
2.142
argmin
0
1
2
3 J
score J, take argmin
(d) Execute
only a_0 of L is applied
p_0
A
B
p_1
$a_0 = (1.5, -1.8)$
$p_1 = (2.5, 1.2)$, swept: free
$a_{1:3}$ not run
$o_1 = R(s_1)$
replan
execute, replan
(e) Candidate action sequences
D Direct
collision
k 0 1 2 3
a_x 2 2 2 2
a_y 0 0 0 0
$E = 16.00$ $C = 1$
$J = 0.1 \cdot 16.00 + 100 \cdot 1 = 101.600$
U Upper detour
k 0 1 2 3
a_x 1.5 2.3 2 2.2
a_y 1.8 0 0.7 −2.5
$E = 26.36$ $C = 0$
$J = 0.1 \cdot 26.36 + 100 \cdot 0 = 2.636$
L Lower detour
selected
k 0 1 2 3
a_x 1.5 2.3 2 2.2
a_y −1.8 0 0.6 1.2
$E = 21.42$ $C = 0$
$J = 0.1 \cdot 21.42 + 100 \cdot 0 = 2.142$
(f) Swept collision test
D, segment $k = 1$: (3, 3) → (5, 3)
A
p_1
free
p_2
free
t
0.00
0.20
0.80
1.00
inside A for $t \in [0.20, 0.80] \Rightarrow C = 1$
agent p_k
goal g
simulated state k
obstacle
simulated path
inside obstacle
executed a_0
$E = \Sigma_k \|a_k\|_2^2$ effort, C collision flag, J cost